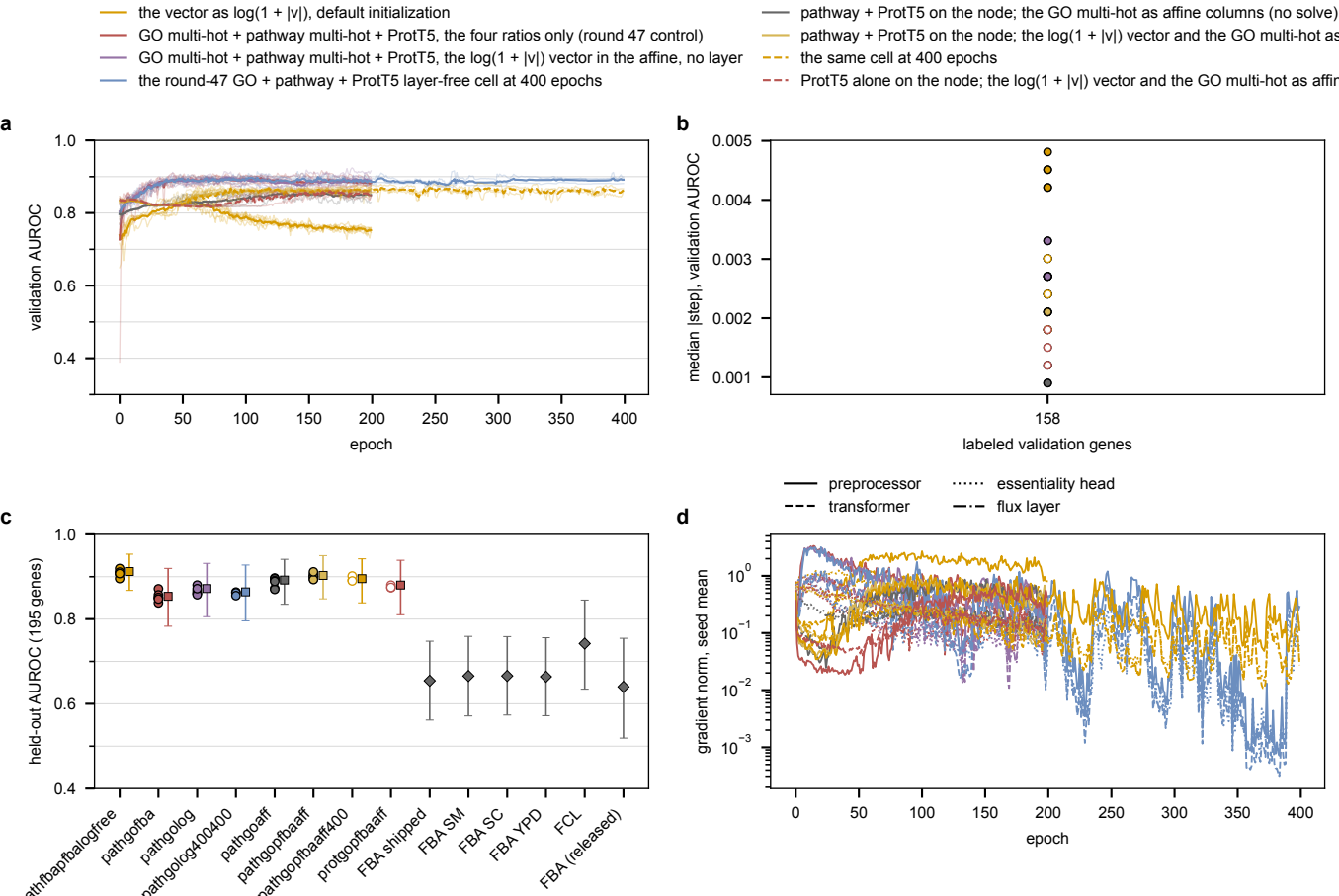

the vector as log(1 + |v|), default initialization
GO multi-hot + pathway multi-hot + ProtT5, the four ratios only (round 47 control)
GO multi-hot + pathway multi-hot + ProtT5, the log(1 + |v|) vector in the affine, no layer
the round-47 GO + pathway + ProtT5 layer-free cell at 400 epochs
pathway + ProtT5 on the node; the GO multi-hot as affine columns (no solve)
pathway + ProtT5 on the node; the log(1 + |v|) vector and the GO multi-hot as
the same cell at 400 epochs
ProtT5 alone on the node; the log(1 + |v|) vector and the GO multi-hot as affir
a
validation AUROC
epoch
b
median |step|, validation AUROC
158
labeled validation genes
preprocessor
transformer
essentiality head
flux layer
c
held-out AUROC (195 genes)
pathgofba
pathgolog
pathgolog400400
pathgoaff
pathgopfbaaff
pathgopfbaaff400
protgopfbaaff
FBA shipped
FBA SM
FBA SC
FBA YPD
FCL
FBA (released)
d
gradient norm, seed mean
epoch